University of Exeter

MTH2008

Real Analysis

On Campus Exam	Closed Book
Additional Materials: None	**Permitted Materials:** No Calculators Permitted

Guidelines:

-The **duration** of this paper is 2hr
-Word Count: No Word Count
-Write all answers in the answer booklet provided.
-Do not open the exam paper until instructed by the Invigilator.
-Mobile and electronic devices are not allowed.
-Do not communicate with other candidates during the exam.

Exam Instructions:

Answer Section A (50%) and any two of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline.

SECTION A

1. (a) (i) State the definition of the upper Riemann sum of a bounded function $f : [0, 1] \to \mathbb{R}$ with respect to a partition $P = (x_0, x_1, \ldots, x_n)$ of $[0, 1]$. (5)

(ii) State Riemann's criterion for integrability of a bounded function $f : [0, 1] \to \mathbb{R}$. (5)

(iii) For $n \in \mathbb{N}$, let $f_n : [0, 1] \to \mathbb{R}$. State sufficient conditions under which

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x = \int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x. \quad (5)$$

(iv) Give an example of a bounded non-Riemann integrable function. (5)

(v) Give a justification for your answer to Part (iv). (5)

(b) (i) State the mean value theorem. (5)

(ii) Define what it means for a function $f : [0, 1] \to \mathbb{R}$ to be Lipschitz continuous. (5)

(iii) Let the function $f : [-1, 1] \to \mathbb{R}$ be defined, for $x \in [-1, 1]$, by

$$f(x) = |x|.$$

Prove that f is Lipschitz continuous but not differentiable. (5)

(iv) Let $f : [0, 1] \to \mathbb{R}$ be a continuous function which is differentiable on $(0, 1)$ with $f(0) = f(1) = 0$. State the conclusion of Rolle's theorem for the function f. (5)

(v) Let $g : [0, 1] \to \mathbb{R}$ denote a continuous function that is differentiable on $(0, 1)$. Further, suppose that there exists $M \in \mathbb{R}$ with

$$|g'(x)| \leq M$$

for all $x \in (0, 1)$. Use the mean value theorem to show that g is Lipschitz continuous. Justify your answer. (5)

[50]

SECTION B

2. (a) State what it means for a function $g : [0, 1] \to \mathbb{R}$ to be differentiable at a point $x \in (0, 1)$, and what it means for g to be differentiable on $(0, 1)$. (5)

   (b) Let $f : \mathbb{R} \to \mathbb{R}$ be defined, for $x \in \mathbb{R}$, by

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

   (i) Prove that f is continuous at zero. (5)

   (ii) Prove that f is differentiable everywhere on its domain, and compute its derivative. (5)

   (iii) Determine if the derivative of f is differentiable. Justify your answer. (10)

   [25]

3. (a) State what it means for a sequence of real numbers $(a_n)_{n\in\mathbb{N}}$ to be Cauchy. (5)

   (b) Let $(a_n)_{n\in\mathbb{N}}$ denote a sequence of real numbers.

   (i) Prove, using Cauchy's convergence criterion for series, that if the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n\to\infty} a_n = 0$. (5)

   (ii) Prove, or give a counterexample to the statement, if $\lim_{n\to\infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ is a convergent series. If giving a counterexample, you may give it without justification. (5)

   (c) Using Cauchy's convergence criterion for series, determine for which values of $p \in \mathbb{R}$, with $p > 0$, the following series is convergent.

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k^p}$$

   Justify your answer. (10)

   [25]

4. (a) State what it means for a function $f : [0,1] \to \mathbb{R}$ to be monotonically increasing. (5)

(b) Prove that if $f : [0,1] \to \mathbb{R}$ is a monotonically increasing function, then f is Riemann integrable. (5)

(c) For $k \in \mathbb{N}$, define $\chi_k : [0,1] \to \mathbb{R}$ by

$$\chi_k(x) = \begin{cases} 0 & \text{if } x \in [0, 2^{-k}), \\ 1 & \text{if } x \in [2^{-k}, 1]. \end{cases}$$

For fixed $n \in \mathbb{N}$, determine if the function $g_n : [0,1] \to \mathbb{R}$ defined, for $x \in [0,1]$, by

$$g_n(x) = \sum_{k=1}^{n} 2^{-k} \chi_k(x)$$

is Riemann integrable. Justify your answer. (5)

(d) For $k \in \mathbb{N}$, let $\chi_k : [0,1] \to \mathbb{R}$ be as defined in Part (c). Determine if the function $g : [0,1] \to \mathbb{R}$ defined, for $x \in [0,1]$, by

$$g(x) = \sum_{k=1}^{\infty} 2^{-k} \chi_k(x)$$

is well defined and Riemann integrable. Justify your answer. (10)

[**25**]

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